Program : **Diploma in Cyber Forensics and Information Security**	
Course Code : **4289**	Course Title: **Cyber Forensics Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To study various forensics tools for forensic investigation.
- Recover deleted files.
- Use of Autopsy for live forensic investigation.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Knowledge in Computer Systems		Introduction to IT Systems Lab	1
Basic Programming Knowledge		Problem Solving and Programming	2
Basic Knowledge in Database		Database Management Systems	3

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Create a Forensic image using FTK Imager	10	Applying
CO2	Implement Data Acquisition, acquisition of cell phones and mobile phones	10	Applying
CO3	Recovering and inspecting deleted files using forensic tools	10	Applying
CO4	Generate report using ProDiscover and FTK	09	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3		1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Create a Forensic image using FTK Imager		
M1.01	Study of Computer Forensics and different tools used for forensic investigation	5	Understanding
M1.02	Create a Forensic image using FTK Imager, create forensic image, check integrity of data, analyze forensic image.	5	Applying
CO2	Implement Data Acquisition, acquisition of cell phones and mobile phones		
M2.01	Perform data Acquisition using USB write Blocker+ FTK imager	5	Applying
M2.02	Acquisition of cell phones and mobile phones	5	Applying
	Lab Test – I	3	
CO3	**Recovering and inspecting deleted files using forensic tools**		
M3.01	Create a malicious document	2	Applying
M3.02	Delete the file	2	Applying
M3.03	Create an image	3	Applying
M3.04	Recover deleted file	3	Applying
CO4	Generate report using ProDiscover and FTK		
M4.01	Create report using ProDiscover	3	Applying
M4.02	Create report using FTK	3	Applying

M4.03	Open ended projects**	3	
	Lab Test – II	3	

****Sample Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation.
Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Study the steps for hiding and extract any text file behind an image file/audio file using command prompt
2. Live forensic case investigation using Autopsy

Text / Reference:

T/R	Book Title/Author
T1	David Benton and Frank Grindstaff ,Practical guide to Computer Forensics - Book Surge Publishing,2006, ISBN-10: 1419623877

Online Resources:

Sl.No	Website Link
1	https://www.geeksforgeeks.org/how-to-create-a-forensic-image-with-ftk-imager/
2	https://medium.com/@tusharcool118/autopsy-tutorial-for-digital-forensics-707ea5d5994d